SDC HACKATHON 2026 | DOMAIN 2

AgriGuard

Team (We got no idea) presents:

AgriGuard



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AgriGuard

Treatment starts days too late

WHY IT MATTERS

Up to 50%
of the crop lost to bacterial leaf blight in conducive conditions
PUBLISHED DATA

RM110M
federal aid to farmers after the 2017 leaf blight outbreak
PUBLISHED DATA

48-72 h
from first symptom to ordering the treatment
TEAM ESTIMATE

WHY IT HAPPENS: FOUR HAND-OFFS, ALL MANUAL

Day 1
Worker spots yellow leaves, WhatsApps a blurry photo

→

Day 2
Owner visits, diagnoses, finds the shed empty

→

Owner then texts the supplier and waits for a reply

→

Day 3-4
Treatment arrives. The disease has already spread

Root cause: diagnosis, stock check and ordering are three separate manual jobs

Sources: J. Current Opinion in Crop Science 2023 (DOA 2019 loss data; RM110M aid in 2017). The 48-72 h delay is our estimate, to be validated in farmer interviews.

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AgriGuard : built for the small farm team

THE SEGMENT (MALAYSIA)

189.5K paddy farmers, our first market
2020 STUDY

45% of individual farm holders are aged 60+
DOSM CENSUS 2024

22.7% of audited paddy farmers earn under RM600 a month
AUDITOR-GENERAL 2022

WHO USES AGRIGUARD

W **Field worker**
Snaps and uploads a photo. Cannot order.

M **Manager**
Checks the diagnosis, overrides if wrong.

O **Owner**
Approves the purchase order from his phone.

Design rule: if you can snap and upload a photo, you can use AgriGuard

Starting with 2-15 person paddy farms, then durian orchards and nurseries. The audit covered about 77,000 farmers in the rice programme. Sources: Agriculture & Food Security 2025; DOSM via NST; Malay Mail.

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AgriGuard connects the whole chain

	WhatsApp + memory	Plant-ID app (Plantix)	Farm software (Agrivi)	AgriGuard
Diagnoses disease from a photo	Owner's eye	Yes	Not listed	Yes
Dosage from the farm's own SOP	No	Generic advice	Not listed	Yes
Checks the farm's chemical stock	No	No	Yes	Yes
Drafts the order, owner approves	No	Links to shops	Manual PO	Yes
Affordable for a smallholder	Free	Free	Quote only	Free tier

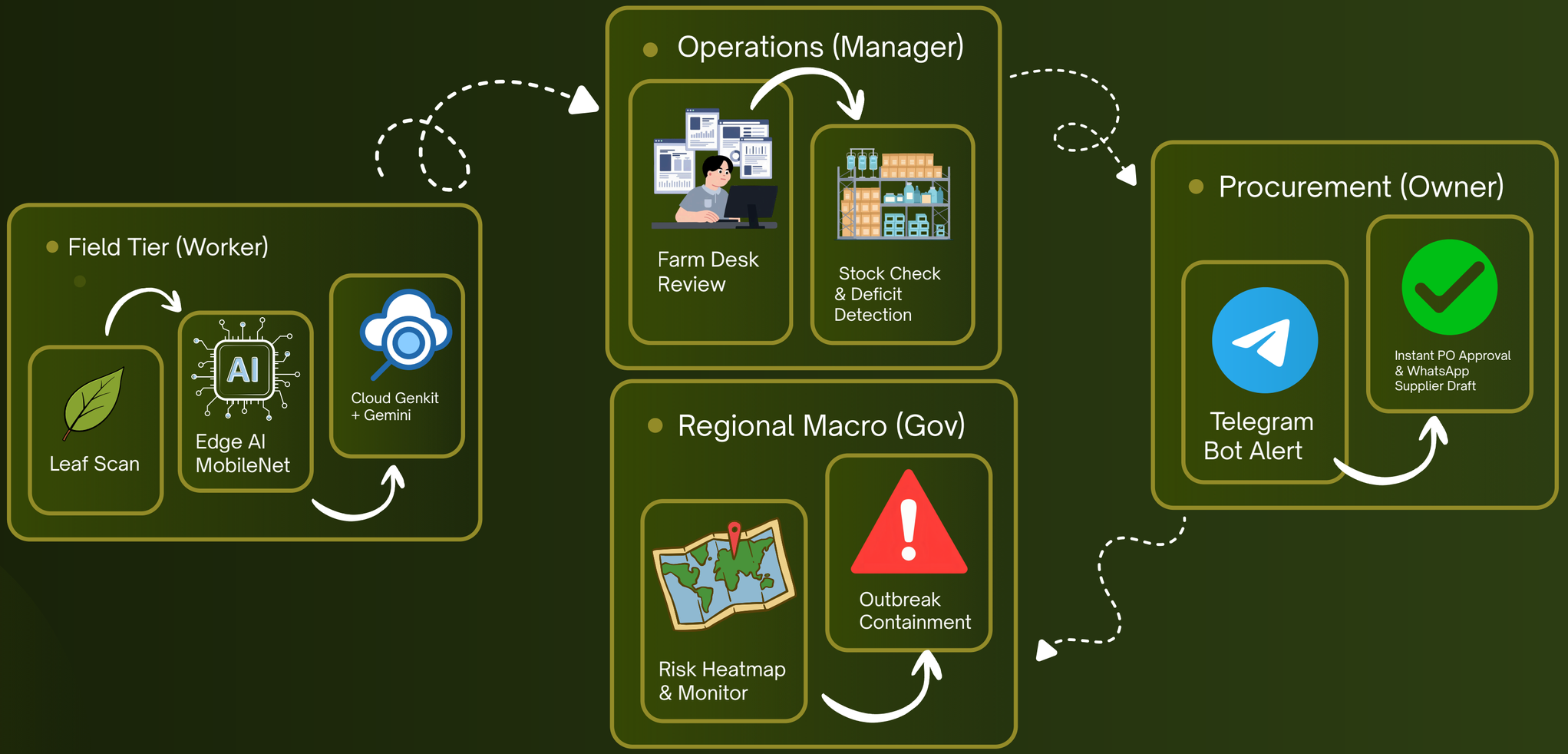
Each option covers one link. **AgriGuard joins diagnosis, stock and an approved order.**

From public descriptions, not hands-on testing: FAO and YourStory 2020 (Plantix links farmers to input shops, in India); Capterra (Agrivi). "Not listed" = absent from those descriptions. AgriGuard = planned MVP scope.

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Solution Workflow & Visual Representation

END-TO-END AGRIGUARD WORKFLOW: FROM CROP TO CO-OP



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Solution Workflow & Visual Representation

END-TO-END AGRIGUARD WORKFLOW: FROM CROP TO CO-OP

Stage 1: Field
Role: Field Worker

- **Instant Screening:** On-device TensorFlow.js with 0ms latency.
- **Coverage:** Detects 80+ plant diseases across 17 crops.

Stage 2: Operations
Role: Operations Manager

- **Scan Triage:** Real-time review of incoming worker scans.
- **SOP Verification:** Curated treatment step confirmation.
- **Stock Check:** Auto-compares required dosage against stock.

Stage 3: Approval
Role: Farm Owner

- **Telegram Bot:** Immediate push notification with itemized cost.
- **1-Tap Action:** Instant Approve / Reject inline buttons.
- **Supplier Dispatch:** Auto WhatsApp PO draft sent to supplier.

Stage 4: Regional
Role: Policy Makers

- **Macro Map:** Anonymized telemetry feeds outbreak heatmap.
- Market Feed: Commodity price & fuel cost correlation.
- GenAI Copilot: Predictive agricultural risk forecasting.

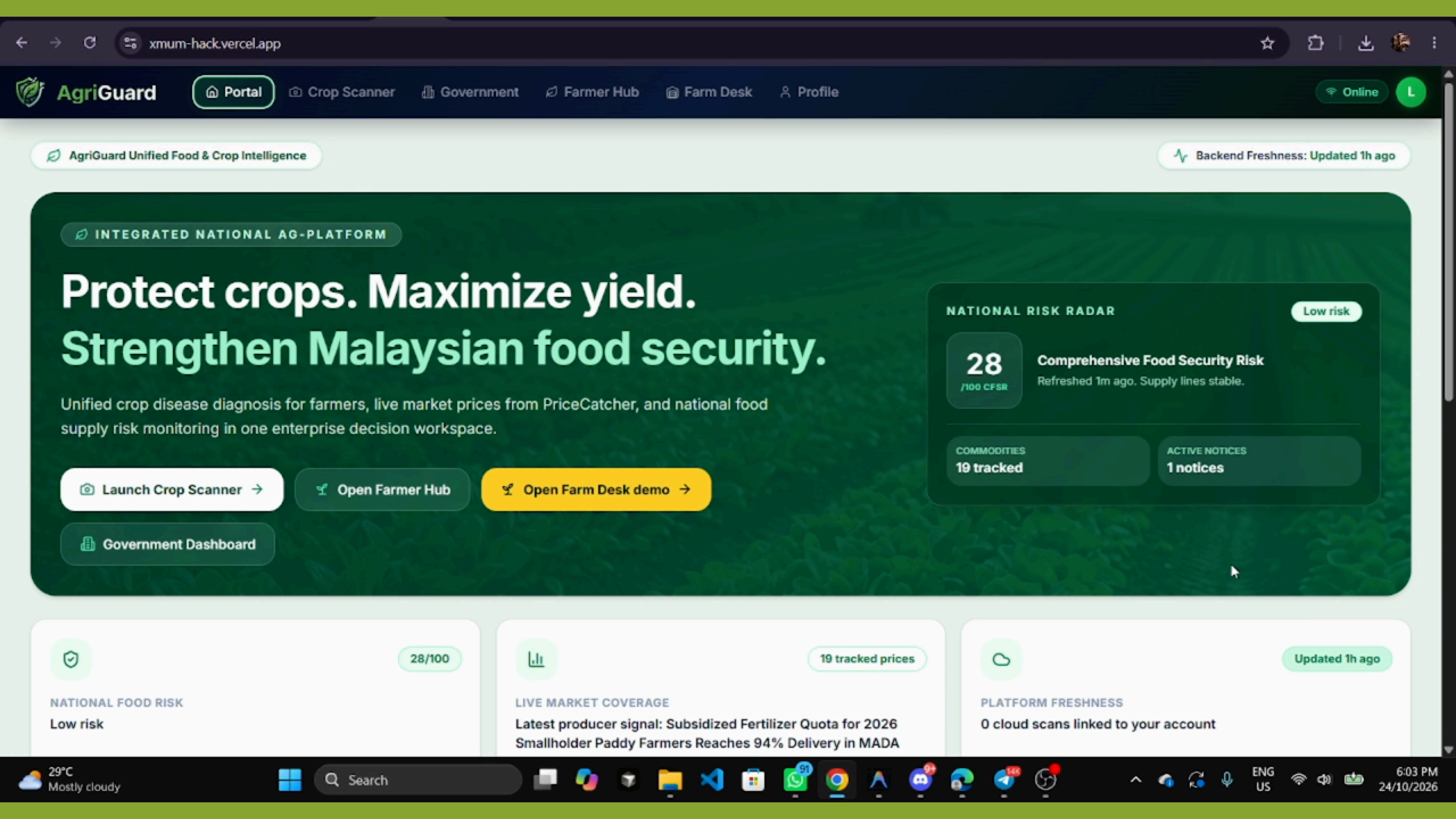
Technology & Architecture

AGRIGUARD STACK



Layer	Technology
Frontend	React 19, TypeScript 5.9, Vite 7, Tailwind CSS 4, Radix UI / shadcn, Recharts, Framer Motion, Wouter
Edge AI	TensorFlow.js 4.22 + MobileNet 2.1.1 (on-device, offline leaf & disease screening)
Backend	Node.js 20+, Express 4.21, tRPC 11.6
Cloud AI	Google Genkit 1.32, Gemini 3.1 Flash Lite, Vertex AI Search (RAG), Pl@ntNet API, Groq vision fallback
Database & Storage	Supabase
Deployment	Vercel (web frontend), Render (Dockerized backend API)
Layer	Technology


AgriGuard
Demo


xmum-hack.vercel.app
AgriGuard
Portal
Crop Scanner
Government
Farmer Hub
Farm Desk
Profile
Online
L
AgriGuard Unified Food & Crop Intelligence
Backend Freshness: Updated 1h ago
INTEGRATED NATIONAL AG-PLATFORM
Protect crops. Maximize yield.
Strengthen Malaysian food security.
Unified crop disease diagnosis for farmers, live market prices from PriceCatcher, and national food supply risk monitoring in one enterprise decision workspace.
Launch Crop Scanner
Open Farmer Hub
Open Farm Desk demo
Government Dashboard
NATIONAL RISK RADAR
Low risk
28
/100 CFSR
Comprehensive Food Security Risk
Refreshed 1m ago. Supply lines stable.
COMMODITIES
19 tracked
ACTIVE NOTICES
1 notices
28/100
NATIONAL FOOD RISK
Low risk
19 tracked prices
LIVE MARKET COVERAGE
Latest producer signal: Subsidized Fertilizer Quota for 2026 Smallholder Paddy Farmers Reaches 94% Delivery in MADA
Updated 1h ago
PLATFORM FRESHNESS
0 cloud scans linked to your account
29°C
Mostly cloudy
Search
ENG
US
6:03 PM
24/10/2026

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IMPACT | WHO BENEFITS AND WHAT CHANGES

FROM FIELD REPORT TO OWNER-REVIEWED ORDER

Why it matters: blight can cause up to 50% crop loss in Malaysia (DOA, cited in J. Curr. Opin. Crop Sci., 2023); a 2025 outbreak in Selangor and Kedah reached 50% incidence (New Disease Reports, 2026).

48 h to 2.4 h

SIMULATED TARGET

Detection to approved order

< 5 min

DEMO TARGET

Photo to draft PO in the live demo

0 auto-orders

BY DESIGN

Owner approves every purchase

3 L needed

WORKED EXAMPLE

2 ha, 1 L in stock, 2 L short: draft PO for one 5 L bottle

Field workers

Report a sick plant in seconds with a photo. No diagnosis skills needed.

Owners and managers

Review one screen and approve a draft PO in minutes. The supplier gets a clear draft order.

Farms and food security

SDG 2: protects small-producer yields and incomes (2.3) and resilience to pests (2.4).

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FEASIBILITY | CAN IT REALISTICALLY BE IMPLEMENTED?

BUILT, RUNNING, READY TO PILOT

Live

Photo to approved PO runs in the demo

Pre-trained

AI models: no training from scratch

Synthetic

No real farm data or live platform needed

Zero

Orders placed without Owner approval

RISK		HANDELED BY
Blurry photo	➡	Backup model + Manager override
Unwanted purchase	➡	Owner approves every purchase order
No signal	➡	Offline scan, syncs later
Wrong dose	➡	Rules set the dose, not the AI

Path to pilot: labelled leaf photos, then one small farm.

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FARMERS START FREE. USAGE PAYS.

Why: nearly 1 in 4 audited paddy farmers earn under RM600 a month, so a fixed subscription would fail. We charge for usage, and suppliers who receive ready orders share the cost.

FARMERS

Free

10 scans a month free, then RM0.50 per scan

SUPPLIERS

3%

of each purchase order (avg RM150 = RM4.50)

GO-TO-MARKET

1. **Pilot**
   100 paddy farms in Kedah and Penang
2. **Partner**
   Granary agencies (MADA, KADA) and agro-chemical suppliers
3. **Expand**
   Durian and nursery templates

Prices are team assumptions, to be validated in farmer and supplier interviews. Source: Auditor-General's Report 2022 (farmer income).

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A RM27M STARTING MARKET

MARKET FUNNEL

1.03M agriculture holders

927K crop holdings

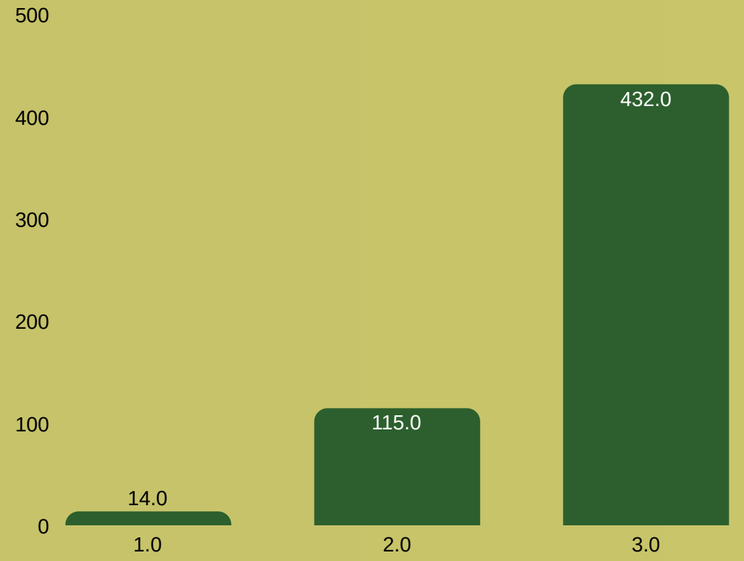
189.5K paddy farmers
= RM27M a year

3,000 farms by Year 3

PER ACTIVE FARM / MONTH

Farm fees	**RM3**
Supplier fees	**RM9**
Revenue	**RM12**
Cost to serve	**~RM2**
Gross margin	**83%**

REVENUE (RM THOUSAND)



Farms: 100 > 800 > 3,000

Break-even at ~150 farms | Farmer pays ~RM36/yr vs RM2-3K lost per outbreak

Sources: DOSM Agriculture Census 2024; Agriculture & Food Security 2025 (2020 count). Team assumptions: 16 scans and 2 POs of RM150 a month, RM1.5K/month infrastructure, RM2-3K outbreak loss on 2 ha.


THANK YOU